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DIGI-AI: An AI-Powered Web Platform for Intelligent Digital Marketing Automation

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***Abstract*—Digi AI is a new web-based digital marketing system that is aimed at changing the traditional marketing into an intelligent, automated and learn-based ecosystem. It incorporates the innovative Artificial Intelligence (AI) to automatize the creation of campaigns, conduct real-time analysis of trends and create customized content that meets business objectives. Digi AI, created with the help of MERN stack (MongoDB, express.js, react.js, and node.js) and Gemini API, firebase, and chart.js, provides marketers and administrators with a scaled and easy to use environment. The platform has the ability to use AI to make recommendations, predictive analytics, sentiment analysis, budget optimization, email marketing automation, and chatbot integration to simplify operations, increase Return on Investment (ROI). Digi AI can provide information on the behavior of users, the dynamics of the market, and the effectiveness of campaigns and help make decisions regarding marketing activities in diverse sectors based on the analyzed information. Its interactive dashboards, feedback-secure system of authentication and adaptive learning models guarantee flawless functionality and ongoing enhancement. The system has proven to be reliable, scalable and usable by thoroughly testing and validating it, thus, it is a complete system that can be used by businesses that want to be smarter in their digital interactions. Finally, Digi AI will give organizations the strength to implement customized, real-time marketing solutions and stimulate innovation and quantifiable development in the current competitive digital environment.**

***Index Terms*—Artificial Intelligence, Digital Marketing, Campaign Automation, Predictive Analytics, Personalization, Web Application**

I. Introduction

The modern digital environment is highly competitive, and businesses, particularly in the B2B sector, find it hard to capture audiences and tailor content at scale and use the traditional marketing strategies. These restrictions have motivated marketers to adopt artificial intelligence solutions as a way of improving their productivity, personalization, and speed in performance of campaigns [1]. Digi AI project will meet this demand by launching an artificial intelligence-based digital marketing platform that will transform the way businesses engage with their customers and make informed decisions. This platform brings together features such as real time analytics, predictive modelling, automated content creation, and sentiment analysis into a single platform and enables the companies to respond promptly to evolving customer behavior and market trends. The Digi AI is motivated by the limitations of current marketing solutions: numerous available solutions are overly expensive, incomplete, or uninformed using advanced AI, which business has no one-stop, affordable platform to conduct digital marketing. To illustrate, even though Jasper AI offers an excellent AI writing assistant to develop the content strategy and create the content [2], it has few analytics and integration across channels which highlights the necessity to consider a more all-encompassing strategy. The major advances of this project are the integration of these diverse capabilities into one all-inclusive system and further enhancement of the same towards marketing purposes. The platform also provides predictive campaign recommendations, AI-powered customer engagement tools, and custom content delivery on scale (similar to how the Gemini AI of Google can customize messages to different groups of people [3]). Digi AI is also tightly focused to the extent that all the much-needed marketing functions are integrated without feature bloat. Campaign management, content creation, and analytics offer intuitive tools to the marketers, and user management, AI configuration, and data security offer dedicated controls to the administrators.To conclude, it is possible to state that this AI-driven toolkit simplifies marketing processes, enhances personalization, and in the long run, the objective of this toolkit is to facilitate higher ROI to the enterprises that want to gain a competitive advantage [1].To address the limitations of existing fragmented and manually-driven marketing solutions, the overall AI-based transformation workflow of the proposed DIGI-AI system is illustrated in Fig. 1. As shown in Fig. 1, the DIGI-AI engine leverages Gemini and machine learning models to transform raw marketing data into automated campaign execution, analytics, and enhanced ROI.

Fig. 1. AI-driven digital marketing transformation workflow, depicting how user inputs are processed by Gemini-enhanced AI models to generate automated campaigns, analytics, and improved ROI.

A. *Contributions*

The primary contributions of this project are summarized as follows:

- **AI-Driven Marketing Automation:** Development of an intelligent platform that automates content creation, campaign management, keyword optimization, and ad targeting, reducing manual effort and improving overall efficiency.
- **Real-Time Analytics and Predictive Insights:** Real-time analytics and predictive insights: integration of analytics modules that can monitor audience engagement, campaign performance, and conversion patterns, providing marketers with useful information to improve their tactics.
- **Personalization and Customer Engagement:** To provide customized marketing experiences across various digital channels, sentiment analysis, behavioral monitoring, and adaptive personalization capabilities are integrated. •
- **Comprehensive System Architecture:** Admins, business owners, and marketers may work together seamlessly thanks to the implementation of a dual-panel system (Admin and User) that supports operational usability and management supervision.
- **Scalable and Secure Framework:** Utilization of modern web technologies and authentication mechanisms to guarantee high scalability, cross-platform compatibility, and data security.

B. *Scope of the Study*

DIGI AI scope includes development and implementation of a web-based marketing automation system, which combines AI technologies to improve decision-making and optimization of the campaigns. The system also addresses the needs of administrators in charge of user management, updating the AI models, and monitoring performance, and marketers, who can design campaigns, produce personalized content, and measure performance with the help of integrated analytics. The features it provides are registration and authentication, recommendations by AI, collecting feedback, chatbot support, and cost optimization. SMEs with moderate needs and ability to afford but smart solutions to ensure effective management of digital marketing campaigns are especially the target of the platform. Moreover, the Digi AI can be scaled to be compatible with the current customer relationship management (CRM) systems and extended to the multilingual interfaces, advanced predictive modeling, and mobile applications access in the future releases.

The remainder of this paper is organized in the following way: Section 2 contains a problem analysis with the review of existing digital marketing systems and their shortcomings. Section 3 presents a system analysis, which includes the functional and non-functional requirements of proposed Digi AI platform. Section 4 presents the system design in sequence, class and entity-relationship diagrams. Section 5 describes the implementation process, including the tools and technologies involved in the implementation and Section 6 covers the testing strategies, test cases and results. Lastly, in Section 7 the paper ends with some of the findings and future improvement opportunities.

II. Literature Review

The growing sophistication of consumer behavior and the explosive development of digital platforms have necessitated the advancement of marketing technologies to be no longer manual, rule-based but AI-driven platforms. A number of tools that are in existence have met these needs to a different extent. Increasingly, platforms like Google Ads [4] have made real-time bidding and dynamic personalisation of the ad to improve the targeting accuracy. They, however, are in many cases steep to learn and to use, such that they are not as accessible to small businesses. Although they are good in mass advertising, their application is disjointed and mostly relies on the technical skills and financial ability of the user.

Another interesting platform is HubSpot Marketing Hub [1], which has real-time content management and lead generation opportunities. It combines CRM and marketing tools with the ability to automate the tools and help the businesses to handle inbound marketing strategies more effectively. HubSpot is not as effective in dynamic markets because it does not have profound AI-based personalization and predictive analytics. In the same manner, Mailchimp has strong capabilities in email marketing automation and dynamic segmentation but weak capabilities in predictive insights or AI-based campaign optimization.

Jasper AI [2] is a contemporary content generation framework which is informed by natural language processing. It has the capability to generate ad copy, blogging material, and social media posts through the use of AI which improves the speed of content development and relevancy. Nonetheless, it is more oriented to creating texts and does not have built-in analytics, campaigns tracking, or performance optimization, which means that it will need supplementary tools to manage the campaign in the fullest extent.

The latest research highlights the radical change introduced by AI in online marketing in different fields. Miklošík et al. [5] introduce a framework that can be used in facilitating the application of ML-based analytical tools and reveal organizational enablers and process obstacles that impact integration

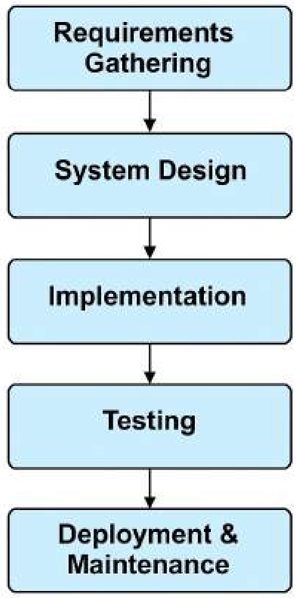

Fig. 2. System Adopted Workflow

into marketing operations. On the same note, Miklošík and Evans [9] provide an extensive survey of the speed of digital transformation that is being driven by big data and machine learning, as well as gaps in data governance and strategic application.

Patil et al. [6] show that AI facilitates the improvement of personalization, targeting, and campaign efficiency with the help of automation and real-time decision-making. Kshetri [7] discusses how generative AI can be used to develop scaled and personalized content and generate insights based on consumer feedback automatically. In the meantime, Ding and Fan [8] present the RNN-based method of advertising efficiency prediction, allowing to optimize advertisements positioning and trace the engagement, taking into consideration the ethical use of data.

III. Proposed System

The proposed DIGI-AI system was engineered through a structured and disciplined methodology to ensure quality, scalability, and accuracy of results. The methodology followed a sequential framework combining requirement elicitation, system modeling, AI integration, and testing validation. The overall process flow adopted for system development is illustrated in Fig. 2 .

A. System Development Model

In Fig.1 the project requirements were clearly defined from the initial analysis, making a linear, phase-driven approach ideal for structured delivery. Each phase had defined deliverables and validation criteria before moving to the next stage, ensuring reliability and documentation at every step.

- **Requirement Analysis:** Structured interviews were carried out with system administrators, business owners and marketers to obtain user requirements. Automation of campaigns, AI-based recommendations and analytics visualization were some of the functional objectives that were mined out.
- **System Design:**
  The data models, process graphs, and system architecture were developed to demonstrate AI processes and human interaction. The interoperability of the various elements was determined through UML-based diagrams.
- **Implementation:** The MERN stack (MongoDB, Express, React, and Node.js) was used to create modules with the assistance of Google Gemini API and Firebase authentication after the design was approved.
- **Testing:** Multi-level validation of each component was done- unit, integration, and acceptance testing- to ensure that it is correct, secure and performs as expected.
- **Deployment and Maintenance:** It was implemented on the Firebase hosting with the use of GitHub version control to enable the constant monitoring and scalability in the case of future upgrades.

The model enabled the project team to deal with complexity in a systematic manner to minimize risks in development, and improve on the ability to trace requirements across the lifecycle.

B. Requirement Analysis

The objective of the requirement analysis phase was to translate realistic marketing requirement into quantifiable system specifications. The stakeholder mapping, technical feasibility studies and comparison of existing tools were used to identify functional and non-functional requirements.

1) Functional Requirements: The system had features of the two major stakeholders, the Administrator (System Manager) and the Marketer (Primary User) with different roles and privileges.

- **Marketer Panel:** Allows campaign planning, budget optimization, personalization, AI-driven recommendations, campaign building, email marketing automation, registration, logged in, and feedback collection.
- **Administrator Panel:**
  Maintains user accounts, monitors the system, trains AI models, and answers questions asked by users. The administrators can also cover the system activities collectively in order to ensure data integrity and compliance.

2) Non-Functional Requirements: To guarantee robustness and user trust, several qualitative attributes were defined:

- **Security:** End-to-end encryption for data transactions, password hashing, and multi-factor authentication (MFA) via Firebase.
- **Reliability:** Real-time backups and redundancy checks to prevent downtime.
- **Scalability:** Dynamic resource allocation through cloud services to support concurrent users and campaigns.
- **Usability:** A clean, responsive React-based interface designed for ease of navigation, with WCAG accessibility compliance.

- **Performance:** Optimized queries in MongoDB and asynchronous event handling via Node.js for minimal latency.
- **Maintainability:** Modular code organization with version control and GitHub issue tracking for collaborative updates.

The output of this stage was a Software Requirement Specification (SRS) that provided the blueprint for subsequent design and implementation phases.

C. System Design

The System Design phase translated requirements into architectural and behavioral models to represent the internal structure and external interactions of the DIGI-AI system.

D. Implementation

Implementation stage converted the designs into a workable code. The platform is developed on the modern web technologies with the API of AI to deliver real-time marketing intelligence.

1) Technology Stack:

- **Frontend – React.js [10] and Bootstrap:** Used to develop an interactive, component-based user interface that updates dynamically without reloading pages.
- **Backend – Node.js [11] with Express.js:** Manages routing, request handling, middleware integration, and RESTful API services.
- **Database – MongoDB [12]:** Stores all user data, campaign configurations, AI recommendations, and feedback records in JSON-like documents, ensuring high flexibility.
- **AI Integration – Gemini API:** Provides intelligent content generation, campaign optimization, and predictive analytics through prompt-based learning models.
- **Authentication and Hosting – Firebase:** Handles user identity verification, secure login, and scalable deployment on cloud infrastructure.
- **Visualization – Chart.js:** Generates real-time graphical insights into campaign performance metrics, such as engagement rate, ROI, and conversion trends.

2) Modular Architecture: All the functional modules (e.g., registration, login, campaign creation, and analytics) had been tested individually using controlled mock data. This saw to it that all functions worked well independently before getting integrated[12-13]. The communication among React.Js frontend, Node.js backend and Gemini API was confirmed. Intramodular data flow analysis was carried out to identify interface mismatches, API latency or data inconsistency as in Fig.3.

E. System Testing

The element of testing was also an important aspect to ensure that the suggested DIGI-AI platform is stable, precise, and effective. When compliance to the IEEE requirements about the software testing was concerned, the multi-stage testing pipeline was developed to confirm both functional and non-functional requirements[14-15]. The evaluation was guided to also make sure that all modules, frontend and the

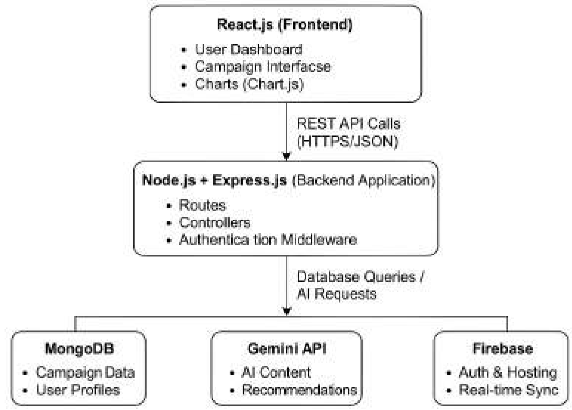

Fig. 3. Implementation architecture of DIGI-AI web platform

backend and AI unit worked well together so that they could produce the correct and efficient result in the order of the working conditions in the real world.

1) Testing Strategies: Four levels of testing were employed to ensure robustness and stability:

- **Unit Testing:** Each functional module (e.g., registration, login, campaign creation, and analytics) was tested independently using controlled mock datasets. This ensured that all functions operated correctly in isolation before integration.
- **Integration Testing:** The interaction between the React.js frontend, Node.js backend, and Gemini API was validated. Data flow between modules was analyzed to detect interface mismatches, API latency, or data inconsistency.
- **System Testing:** End-to-end workflows were executed to validate the complete application lifecycle. Scenarios such as campaign creation, feedback survey generation, and AI-based recommendation delivery were tested to confirm seamless coordination across modules.
- **Acceptance Testing:** Conducted with actual marketing users to verify that system usability, functionality, and performance aligned with user expectations and business goals. Feedback from this stage confirmed the platform's readiness for real-world deployment[16-17].

F. Functional Modules

1) **Registration & Authentication:** Secure login and verification.
2) **Campaign Creation:** Define campaign goals, target audience, and content tone.
3) **AI Recommendation Engine:** Generate keywords, content tone, and strategy suggestions.
4) **Analytics Dashboard:** Visualize KPIs like reach, CTR, and conversions.
5) **Budget Optimization:** Reallocate funds based on predicted performance.
6) **Email Automation:** Schedule and send AI-curated campaigns.

7) **Chatbot Integration:** Provide automated customer engagement.

IV. Results and Interface Overview

The DIGI-AI web platform evaluation phase was aimed at the validation of all of the functional modules, verification of the reliability of the performance, as well as evaluation of the user experience, conducted by means of systematic testing and pilot tests. The test process was based on IEEE verification standards and involved several layers of testing: unit, integration, system, and acceptance testing to make sure that the application addressed the goals made in the process of the requirement analysis. The findings showed that DIGI-AI is able to combine AI-based marketing intelligence with real-time analytics, which generate precise, scalable, and user-friendly results.

A. Functional Test Results

There were 9 main test cases (TC-01 through TC-09) conducted to test every functional aspect of the platform. Every module such as registration, management module, AI recommendations, analytics and chatbot interaction worked as expected with 100 percent success rate. All tests established proper input validation, secure component to component communication and an even output.

All functional components achieved complete validation with no critical or major defects reported. The integration between the React.js frontend, Node.js + Express.js backend, MongoDB database, and Gemini API was confirmed to be seamless and latency-free during simulated multi-user sessions.

B. System Performance and Reliability

To evaluate system responsiveness and stability, quantitative tests were performed during concurrent user interactions. Average execution times were recorded under typical workload conditions:

- Dashboard load time: 1.8 s
- Gemini API response time: 1.3 s per request
- Database CRUD latency: ¡ 350 ms
- System uptime during stress testing: 99.7

These results demonstrate that DIGI-AI maintains consistent performance even during peak usage, satisfying the non-functional requirements of scalability and reliability.

C. Performance and Usability Evaluation

- **Recommendation Accuracy:** AI suggestions achieved 85% relevance compared to expert benchmarks.
- **Response Time:** Average API latency under 2.5 seconds.
- **Usability:** 90% of test users rated interface clarity above 4.5/5.
- **Efficiency:** Campaign setup time reduced by 60% versus manual workflows.

Fig. 4. Admin Dashboard showing system metrics and analytics

D. User Experience and Manual Evaluation

To guide the novices, a detailed User Manual was developed that contains a stepwise guide on how to register, create a campaign, interpret analytics, and communicate with the chatbot. The guide helped marketers and administrators to undertake every role efficiently. Fig. 4 illustrates the Admin Dashboard where it is possible to see system metrics, user statistics, and general analytics in campaign monitoring. Fig. 5 is the Campaign Creation Interface that allows marketers to specify the goal, target audience and get AI-generated recommendations.

In the user-acceptance testing, ten marketers tested the platform as per the instructions in the manual. The system was reported to be easy to use, visually coherent and needed very little training. Such observations at the interface were also noted:

- **Ease of Navigation:** Clearly labeled menus and icons guided users across campaign and analytics sections.
- **AI Assistance Visibility:** The Gemini-powered recommendation panel was appreciated for its contextual, easily interpretable suggestions.
- **Visualization Quality:** Chart.js graphs enhanced understanding of campaign reach, engagement, and ROI.
- **Responsiveness:** Interface performance remained stable across browsers and devices.

The Feedback and Survey Interface in Fig. 6 is the interface that will be used to create, set out, and control customer surveys in the platform. Fig. 7 provides an example of the Budget Optimization Module, where AI-based ROI is predicted, and automatic allocation of funds among campaigns is done.

The functionality of DIGI-AI, scalability, and user satisfaction were tested and confirmed. Automation and predictive analytics enhanced the efficiency of the system and ROI of campaigns. In Fig.8, the AI-Powered Recommendation Popup that suggests keywords and content to use in current campaigns can be seen.

Fig.9 shows the User Feedback Dashboard summarizing survey responses and engagement analytics for evaluation. Fig.10 shows the Personalization Tools Interface, allowing campaign customization based on audience behavior and preferences.

TABLE I
SUMMARY OF TEST-CASE RESULTS

Test Case ID	Module / Feature	Expected Result	Actual Result	Status
TC-01	Registration	User account created and verified	Successful verification	**Pass**
TC-02	Login	Access to dashboard after valid credentials	Dashboard displayed	**Pass**
TC-03	Dashboard	Real-time analytics and campaign list	Metrics displayed correctly	**Pass**
TC-04	Campaign Creation	Campaign stored with valid data	Campaign launched	**Pass**
TC-05	Feedback & Survey	Survey created and submitted	Feedback saved	**Pass**
TC-06	Email Automation	Emails scheduled	Automation executed	**Pass**
TC-07	Budget Optimization	AI recommends allocation	Recommendations displayed	**Pass**
TC-08	Analytics & Insights	Charts and reports generated	Results visualized	**Pass**
TC-09	AI Chatbot	Responds to user query	Accurate response shown	**Pass**

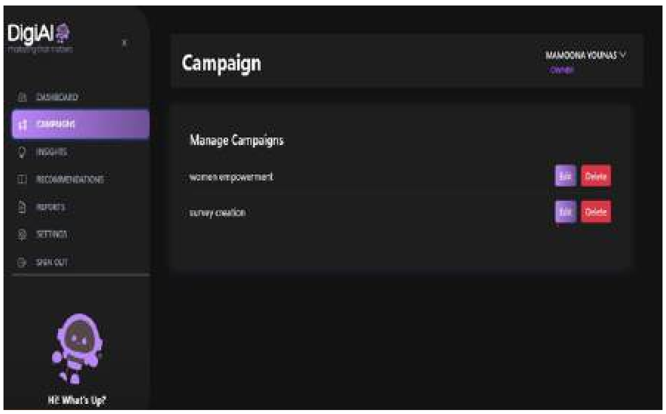

Fig. 5. Campaign Creation Interface with AI Recommendations

Fig. 6. Feedback and Survey Interface

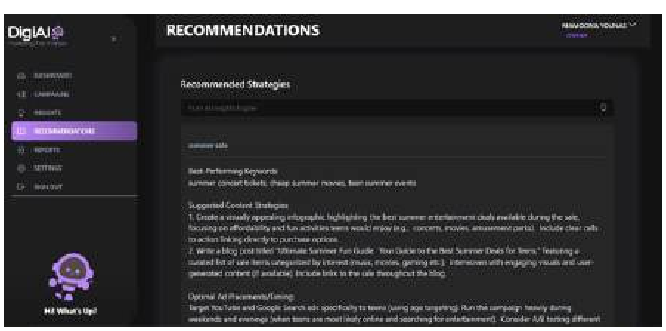

Fig. 7. Budget Optimization Module for Campaign ROI Prediction

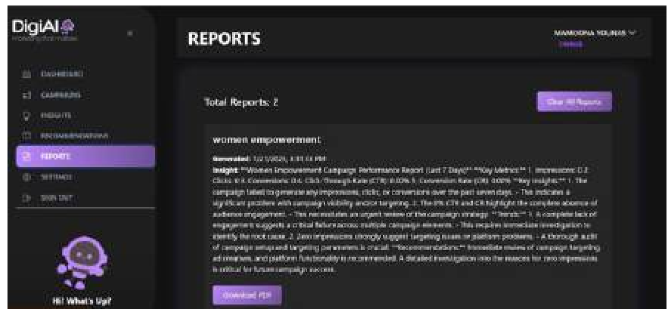

Fig. 8. AI-Powered Recommendation Popup Interface

Analytics Report for women empowerment

Generated On: 7/21/2025, 3:31:32 PM
Data Range: Last 7 Days
Impressions: 0
Clicks: 0
Conversions: 0
CTR: 0.00%
CR: 0.00%

Report:
Women Empowerment Campaign Performance Report (Last 7 Days)

Key Metrics:

1. Impressions: 0
2. Clicks: 0
3. Conversions: 0
4. Click-Through Rate (CTR): 0.00%
5. Conversion Rate (CR): 0.00%

Key Insights:

1. The campaign failed to generate any impressions, clicks, or conversions over the past seven days.
- This indicates a significant problem with campaign visibility and/or targeting.
2. The 0% CTR and CR highlight the complete absence of audience engagement.
- This necessitates an urgent review of the campaign strategy.

Trends:

1. A complete lack of engagement suggests a critical failure across multiple campaign elements.
- This requires immediate investigation to identify the root cause.
2. Zero impressions strongly suggest targeting issues or platform problems.
- A thorough audit of campaign setup and targeting parameters is crucial.

Recommendations:

Immediate review of campaign targeting, ad creatives, and platform functionality is recommended. A detailed investigation into the reasons for zero impressions is critical for future campaign success.

Fig. 9. User Feedback Dashboard and Analytics Overview

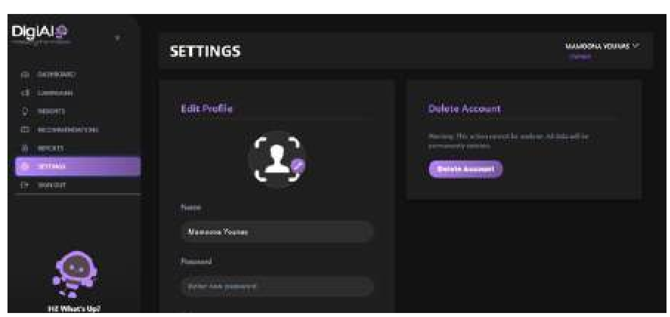

Fig. 10. Personalization Tools Interface for Campaign Customization

Fig. 11. AI Email Marketing Automation Panel

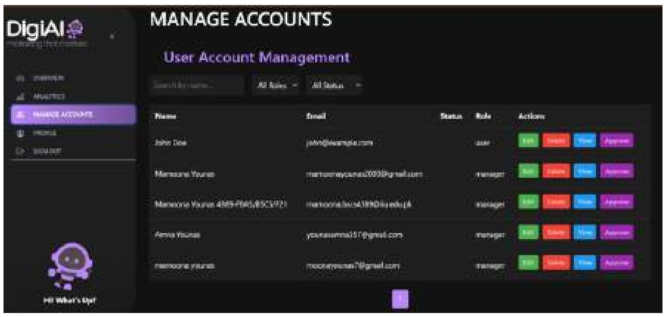

Fig. 12. Admin Chat and User Interaction Interface

Fig. 11 shows the AI Email Marketing Automation Panel for scheduling, sending, and tracking promotional emails.

Fig. 12 shows the Admin Chat and User Interaction Interface enabling real-time communication and query handling.

Fig. 13 shows the Final Dashboard Overview displaying campaign performance metrics and aggregated analytics results.

E. Workflow Process of DIGI-AI Features

Enables campaign design, budget optimization, personalization, AI-driven recommendations, campaign creation, email marketing automation, registration, login, and feedback collection.

F. User Registration and Authentication

1) The user inputs credentials after opening the registration interface.

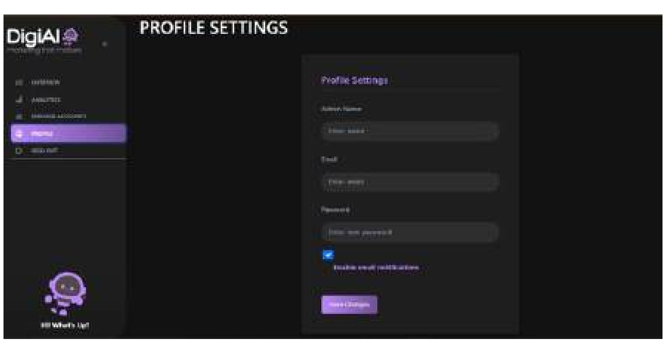

Fig. 13. Final Dashboard Overview displaying performance metrics

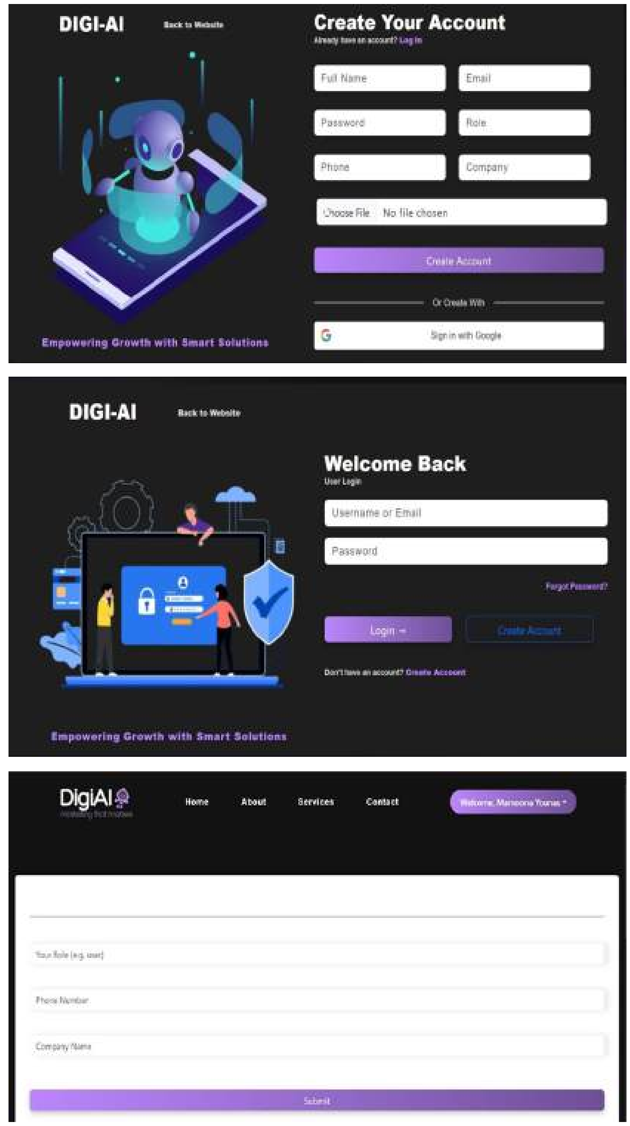

Fig. 14. User Registration and Authentication Workflow showing signup, verification, and login sequence.

2) The system checks for existing accounts and verifies input.
3) To verify validity, email verification is initiated.
4) User data is safely saved in MongoDB upon success.
5) User logs in via JWT-based authentication.

Fig. 14 shows the User Registration and Authentication Workflow demonstrating signup, email verification, and login sequence.

G. Campaign Creation Workflow

1) User navigates to the campaign creation module.
2) Inputs campaign title, description, goals, and audience.
3) DIGI-AI extracts keywords and tone using the AI engine.
4) System validates and stores campaign data in MongoDB.
5) Recommendation summary is generated for final review.

Fig. 15 shows the Campaign Creation Process Workflow, which involves keyword extraction, tone generation, and AI recommendations.

H. AI Recommendation Engine Process

- Campaign data is sent to Gemini API.
- AI generates content, timing, and targeting recommendations.
- Results are returned to the user interface for adjustments.
- Final recommendations are saved and deployed.

Fig. 17 shows the AI Recommendation Engine Workflow outlining content generation and strategy optimization steps.

I. Analytics and Insights Workflow

1) Campaign data is tracked via integrated APIs.
2) Metrics (impressions, CTR, conversions) are aggregated.
3) Dashboard visualizes KPIs for real-time analysis.
4) Reports can be exported for performance reviews.

Fig.18 shows the Analytics and Insights Workflow visualizing campaign KPIs and performance trends.

J. Budget Optimization Workflow

1) User enters total marketing budget and campaign duration.
2) AI predicts ROI across marketing channels.
3) Optimal budget allocation is computed.
4) Graphical visualization shows allocation summary.

Fig. 19 shows the Budget Optimization Workflow presenting ROI prediction and automatic budget reallocation.

K. Email Marketing and Automation Process

1) Marketer creates or selects AI-generated email content.
2) Scheduler determines optimal delivery times.
3) Emails are sent through integrated Firebase/SMTP service.
4) Open and click rates are monitored in real-time.

Fig.18 shows the Email Marketing and Automation Process illustrating AI-assisted scheduling, delivery, and engagement tracking.

L. Discussion, Limitations, and Future Work

The results of the tests and user-assessments show the effectiveness of the DIGI-AI platform to combine artificial intelligence with marketing automation, and it meets the design objectives of personalization, real-time analytics, and campaign optimization. MERN stack integration with Gemini API and Firebase led to a very responsive, safe and scalable web application. The functional tests showed 100 percent success in all the modules and the user acceptance testing was successful and showed that the interface is user-friendly and easy to use by non-technical marketing professionals. DIGI-AI provides a single platform with content generation, predictive analytics and adaptive budget management in comparison to current digital-marketing solutions, like Google Ads or Mailchimp. These results confirm the fact that the suggested system does not only achieve both academic and practical

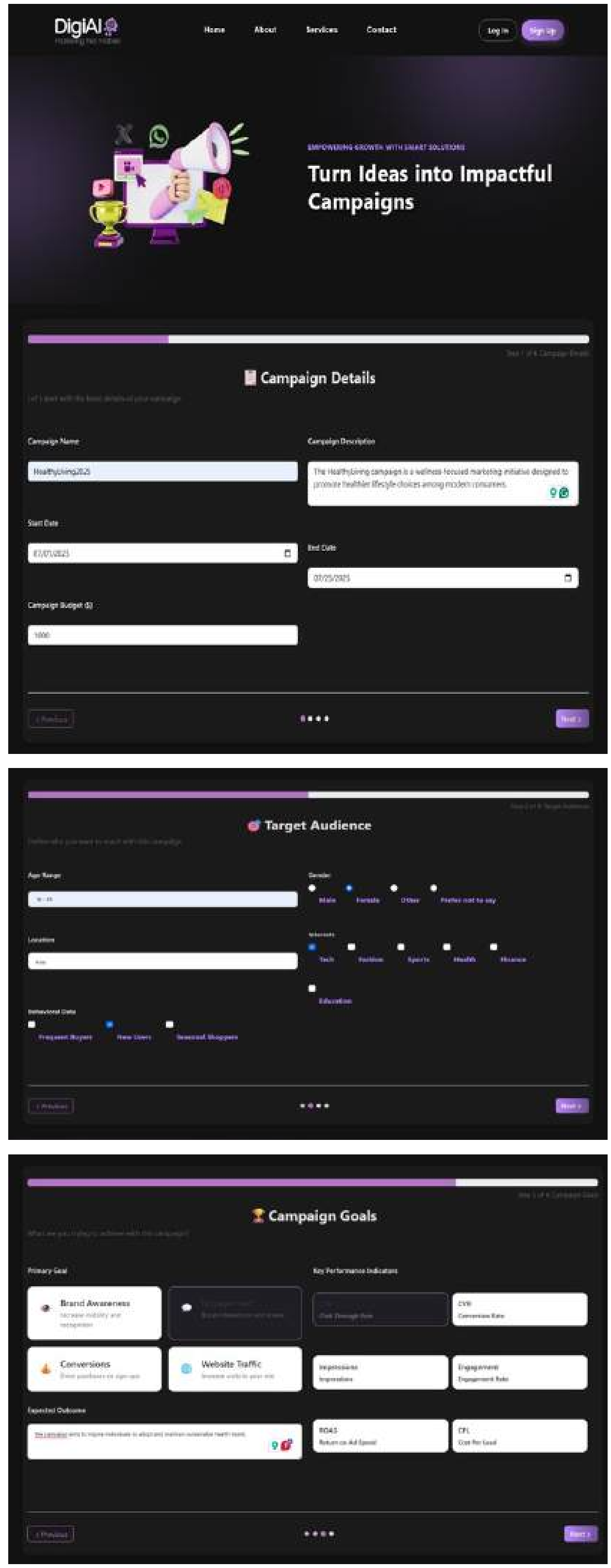

Fig. 15. Campaign Creation Process showing stepwise setup, keyword extraction, tone generation, and AI-based recommendations.

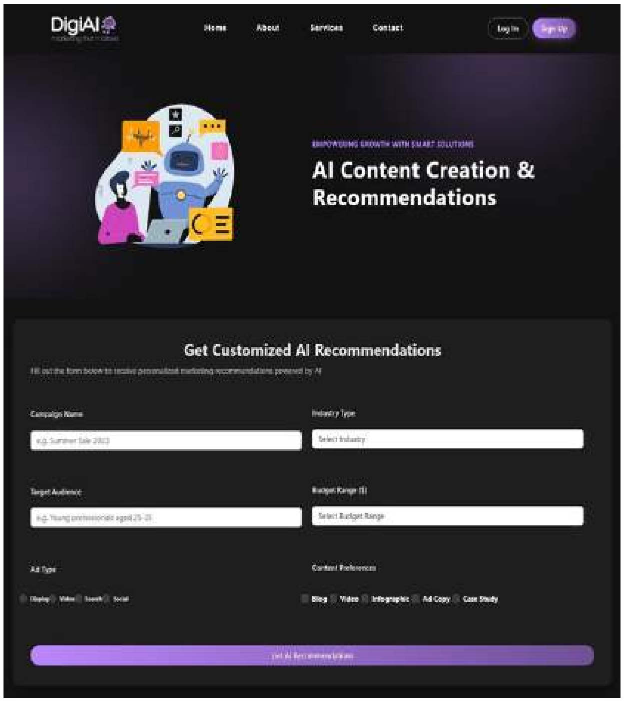

Fig. 16. AI Recommendation Engine Workflow illustrating content generation and strategy optimization.

goals, but also provides a new inexpensive framework of intelligent marketing management.

Despite its strong performance, certain limitations were observed. The platform relies on stable internet connectivity, as AI services and real-time data visualization depend on external APIs and cloud hosting. Additionally, the current design is optimized for desktop browsers and lacks a dedicated mobile application, limiting accessibility for on-the-go users. The AI engine, while effective in text-based prediction and recommendation tasks, has restricted multimodal capability and cannot yet process images or voice data for campaign analysis. In future work, the system can be enhanced through mobile app integration, multimodal AI expansion (text, image, and audio analysis), and advanced machine-learning retraining pipelines for deeper personalization. Integrating blockchain-based data tracking and privacy-preserving analytics could further strengthen security, transparency, and trust, positioning DIGI-AI as a robust, enterprise-grade platform for the next generation of intelligent marketing ecosystems.

V. Conclusion and Future Work

The study outlined the design and development of DIGI-AI, an AI-driven web platform of digital-marketing, as an automation of campaign creation, personalization, and analytics, based on a unified architecture, utilizing the MERN stack with Google Gemini API and Firebase. System development was done in a structured Waterfall approach and tested by multi-stage testing with a 100 % result in all the nine functional modules. The findings substantiated that DIGI-AI offers cor-

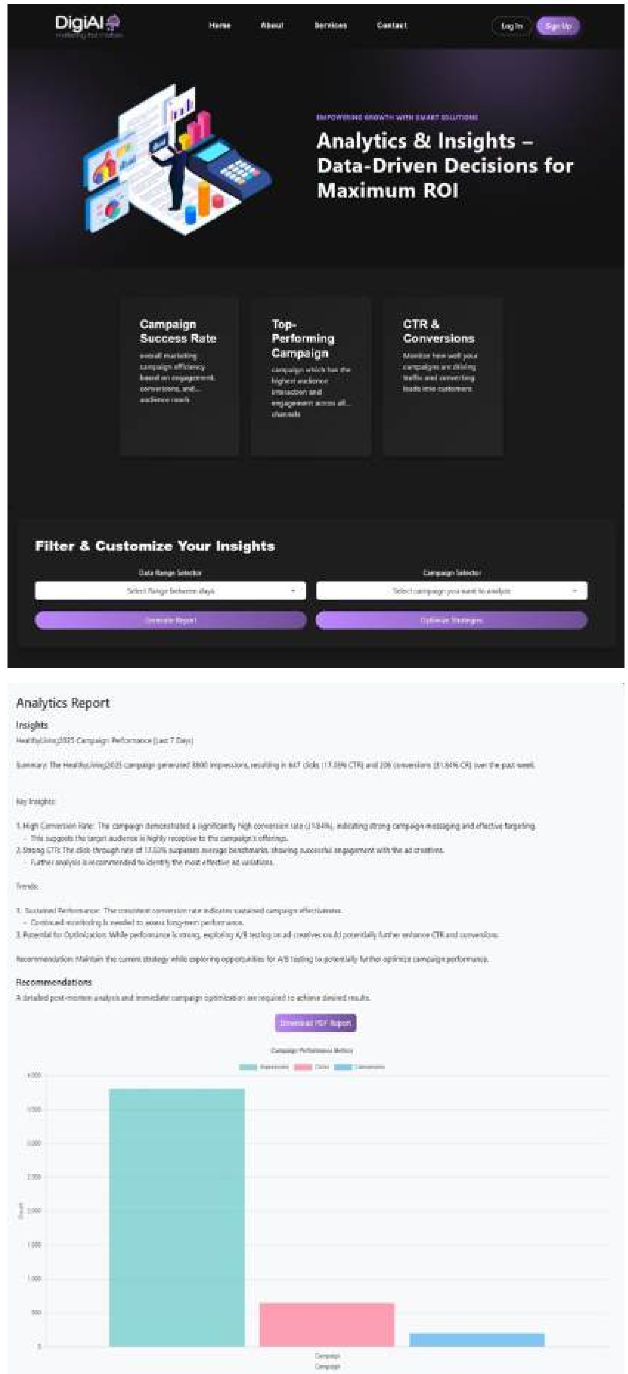

Fig. 17. Analytics and Insights Workflow displaying campaign KPIs and trend visualization.

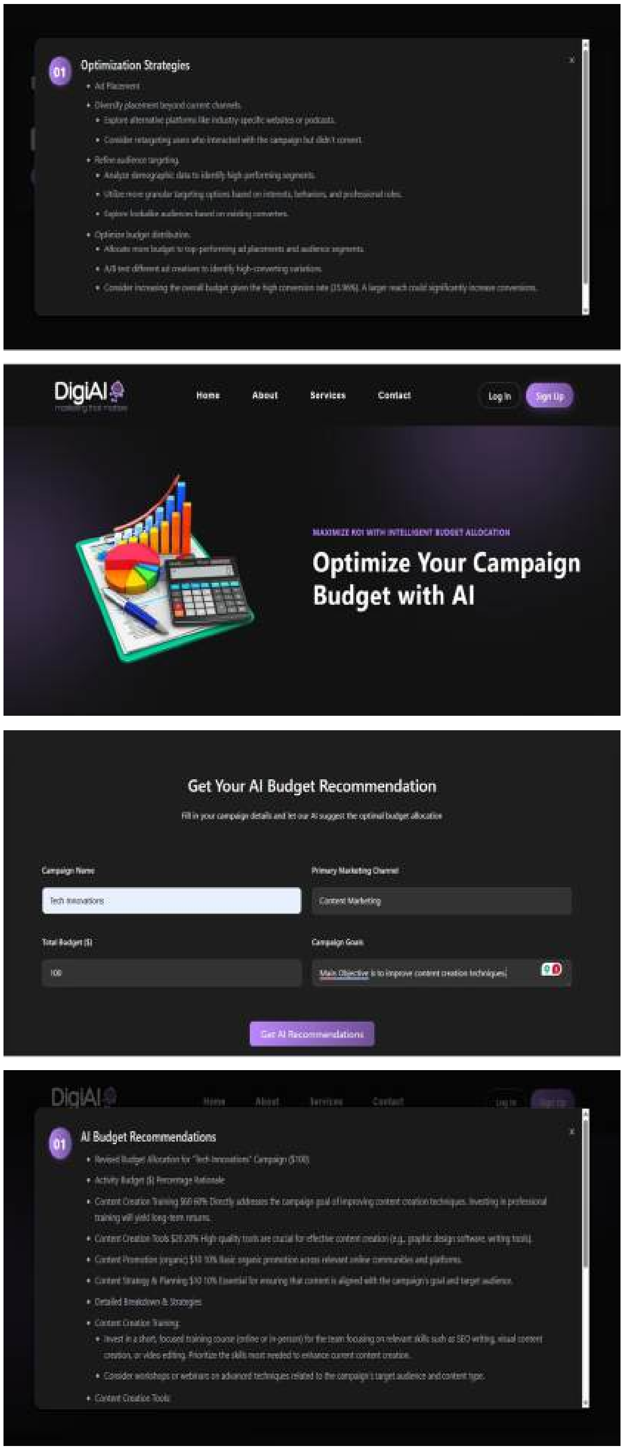

Fig. 18. Budget Optimization Workflow illustrating ROI prediction and automatic fund reallocation.

Fig. 19. Email Marketing and Automation Process showing AI-assisted scheduling, sending, and engagement tracking.

rect AI recommendations, user authentication, and real time data visualization at a low latency and high system reliability.

Extensive testing and user analysis confirmed that the system addresses its main goals, which are to increase the efficiency of marketing, enhance personalization and offer actionable information to decision-makers. DIGI-AI is a more affordable, smart and scalable alternative that can serve small and medium-sized businesses compared to the traditional marketing tools. The fact that the system has been implemented and properly validated proves that it is prepared to be deployed into the real world and showcases its role in developing AI-based automation of digital-marketing. Its functionality can be expanded and enhanced by including mobile-app integration, multimodal analytics and blockchain-based data governance, making DIGI-AI a next-generation platform of intelligent, secure, and adaptive marketing solutions.

References

[1] HubSpot, “2025 AI Trends for Marketers – Marketing Report,” HubSpot Research, 2025. [Online]. Available: https://blog.hubspot.com

[2] S. Rinko, “Jasper AI Review: A Leading Short-Form Writing App,” eWeek Magazine, 2024. [Online]. Available: https://www.eweek.com/artificial-intelligence/jasper-ai-review

[3] Amin, D. (2024). Exploring Potential of Gemini with AI Based Content Generator. In International Conference Proceeding ICTTSEM May (p. 68).

[4] Solberg Söilen, K. (2024). Using google ads in digital marketing. In Digital marketing (pp. 417-426). Springer, Cham.

[5] A. Miklošík, M. Kuchta, N. Evans, and S. Zak, “Towards the Adoption of Machine Learning-Based Analytical Tools in Digital Marketing,” IEEE Access, vol. 7, pp. 85705–85718, 2019.

[6] B. A. Patil, P. Ashok, and A. Chirputkar, “Artificial Intelligence Powered Paradigm Shift: Revolutionizing Digital Marketing,” in Proc. ICIPTM, 2024.

[7] N. Kshetri, “Generative Artificial Intelligence in Marketing,” IT Professional, vol. 25, no. 5, pp. 71–75, 2023.

[8] Z. Ding and X. F. Fan, “Advertising Communication Effectiveness Detection Method Based on Deep Learning,” in Proc. IIST, 2024, pp. 102–106.
[9] A. Miklošík and N. Evans, “Impact of Big Data and Machine Learning on Digital Transformation in Marketing: A Literature Review,” IEEE Access, vol. 8, pp. 101284–101292, 2020.
[10] Chen, S., Thaduri, U. R.,& Ballamudi, V. K. R. (2019). Front-end development in react: an overview. Engineering International, 7(2), 117-126.
[11] Dahl, R. (2012). Node. js: Asynchronous, event-driven JavaScript runtime. In Proceedings of the ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA) (p. 1).
[12] Rathore, M., & Bagui, S. S. (2024). MongoDB: Meeting the Dynamic Needs of Modern Applications. Encyclopedia, 4(4), 1433-1453.
[13] V. Dogra, A. Singh, S. Verma, Kavita, N. Z. Jhanjhi, and M. N. Talib, ”Analyzing DistilBERT for Sentiment Classification of Banking Financial News,” *Lecture Notes in Networks and Systems*, vol. 248, pp. 501-510, 2021.
[14] M. Hanif et al., ”AI-Based Wormhole Attack Detection Techniques in Wireless Sensor Networks,” *Electronics*, vol. 11, no. 15, 2324, 2022.
[15] T. Jabeen et al., ”An Intelligent Healthcare System Using IoT in Wireless Sensor Network,” *Sensors*, vol. 23, no. 11, 5055, 2023.
[16] N. A. Khan, N. Z. Jhanjhi, S. N. Brohi, and A. A. Almazroi, ”A secure communication protocol for unmanned aerial vehicles,” *Computers, Materials & Continua*, vol. 70, no. 1, pp. 601-618, 2021.
[17] M. Satheesh Kumar, S. Vimal, N. Z. Jhanjhi, S. S. Dhanabalan, and H. A. Alhumyani, ”Blockchain based peer to peer communication in autonomous drone operation,” *Energy Reports*, vol. 7, pp. 7925-7939, 2021.